
EFFECT OF BACKGROUND MUSIC TYPE AND TIME AWAKE ON TYPING PERFORMANCE

A PREPRINT

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ABSTRACT

Background music is often promoted as a productivity aid, yet empirical evidence for its effectiveness during routine computer work is mixed. Likewise, the amount of time one has been awake is known to influence vigilance, but its interaction with music remains unexplored. This pilot study therefore examined the combined effect of *music tempo* (silence, classical ~ 60 – 80 BPM, hardstyle ~ 150 BPM) and *time awake* (within 1 h, 4–8 h, ≥ 12 h since waking) on typing performance.

Neither main nor interaction effects of music or time awake reached statistical significance for speed or accuracy. There seems to be a performance peak at 4–8 h awake regardless of music condition.

1 Introduction

Ambient music is normal in modern study and office environments. Some commercial playlists promise increased focus, yet laboratory research offers contradictory results: classical music like Mozart can aid with cognitive performance [Angel et al., 2010], fast tempo may increase arousal but also distraction [Salamé and Baddeley, 1989]. In parallel, the neurocognitive literature shows a clear pattern in attention and motor speed—vigilance improves for several hours after waking and declines after prolonged wakefulness [Borbély, 1982].

This present experiment tests whether typing speed and accuracy depend simultaneously on background-music tempo and time awake. Typing is an everyday motor-cognitive task, making the findings relevant.

The objective and hypothesis for this experiment is to test whether typing performance is affected by both background music, and time awake. We propose the following hypothesis:

Null hypothesis (H_0): No main or interaction effects of music or time awake on typing performance.

Alternative hypothesis (H_1):

- (a) Classical music (~ 60 – 80 BPM) improves typing speed and accuracy relative to silence and hardstyle (~ 150 BPM).
- (b) Performance peaks after 4–8 hours of wakefulness.
- (c) Favorable music mitigates performance decline during late-evening conditions.

2 Methodology

2.1 Design and factors

The study followed a 3×3 full-factorial, repeated-measures design with two fixed factors:

- **Music type:** *Silence*, *Classical* (60–80 BPM), *Hardstyle* (~ 150 BPM). Three instrumental tracks per category were rotated to minimise song-specific effects.

- **Time awake:**

1. Within the first hour after waking
2. 4–8 hours awake
3. 12 or more hours awake

Each participant was asked to complete nine one-minute typing sessions (three music conditions $\times$ three time-awake conditions) per day, and encouraged to repeat the experiment as many times as possible over the course of one week.

2.2 Participants

Thirteen volunteers, a mix of family and friends, were gathered for the test.

2.3 Stimuli and Measures

The stimulus set comprised the 200 most common English words [noa], presented in random order *with replacement* for each test.

Typing speed was computed as words per minute (WPM):

$$\text{WPM} = \frac{\text{keystrokes}}{5} \quad (1)$$

Typing accuracy was computed as the proportion of correct actions:

$$\text{Accuracy} = \frac{\text{Number of correct actions}}{\text{Total actions}} \quad (2)$$

where the number of correct actions was defined as the number of typed characters minus the Levenshtein distance [Levenshtein, 1966] between the reference and typed text (up to the length of the typed text). The total actions were the number of keystrokes (tracked, and logged, by the web page) accounting for corrections.

2.4 Data cleaning

A webpage design bug occasionally introduced low-accuracy results. Therefore, rows of data with accuracy $\leq 50\%$ were removed, leading to the removal of eight out of 94 rows. Also, additional outliers were removed using the IQR method.

2.5 Statistical analysis

Analyses were conducted in Python using `statsmodels`, `Pandas`, `Plotly` and more. For each dependent variable a linear mixed-effects model was fitted with *music type*, *time awake*, and their interaction as fixed effects and a random intercept for *participant*. Tukey HSD post-hoc comparisons were performed on the resulting cell means for descriptive completeness. The α level was set to .05.

3 Results

Table 1 summarises the descriptive statistics for both outcomes. Performance peaked consistently in the 4–8 h awake condition, but the variance was large.

Mixed-effects models. The fixed-effects results are reported in Table 2. Neither music type nor time awake, nor their interaction, reached statistical significance for typing speed.

Post-hoc contrasts. Tukey HSD comparisons (Tables 3–4) showed no adjusted p -values below .05, corroborating the global tests.

Visual inspection. Figures 1–6 illustrate the data distribution. The interaction plots display a mild inverted-U pattern over the day, but the confidence bands overlap substantially. Notably, as seen in Figure 6, the Hardstyle condition showed a larger increase in accuracy at 4–8 hours awake compared to Silence and Classical, but also a sharper decrease after 12+ hours awake. The lines are not parallel, suggesting a possible interaction between music type and time awake. However, statistical analysis did not reveal any significant main or interaction effects, as seen in Table 2.

Table 1: Descriptive statistics (mean $\pm$ SD) for typing speed (WPM) and accuracy by music type and time awake.

Music Type	Time Awake	WPM (Mean $\pm$ SD)	Accuracy (Mean $\pm$ SD)
Silence	1h	64.58 $\pm$ 15.84	0.900 $\pm$ 0.050
	4-8h	69.69 $\pm$ 30.03	0.910 $\pm$ 0.050
	12h+	50.26 $\pm$ 16.38	0.870 $\pm$ 0.050
Classical	1h	63.78 $\pm$ 18.65	0.890 $\pm$ 0.040
	4-8h	71.89 $\pm$ 26.15	0.900 $\pm$ 0.050
	12h+	51.00 $\pm$ 17.91	0.870 $\pm$ 0.060
Hardstyle	1h	64.69 $\pm$ 19.11	0.870 $\pm$ 0.040
	4-8h	72.53 $\pm$ 25.46	0.930 $\pm$ 0.040
	12h+	53.36 $\pm$ 19.87	0.860 $\pm$ 0.050

Table 2: Mixed-effects model results for typing speed (WPM) and typing accuracy. Reference group: Silence, 1h awake.

Predictor	Typing Speed (WPM)			Typing Accuracy		
	Coef.	SE	<i>p</i>	Coef.	SE	<i>p</i>
Intercept	54.68	7.09	<0.001	0.910	0.018	<0.001
Classical	-0.80	1.81	0.659	-0.012	0.019	0.514
Hardstyle	0.11	1.81	0.951	-0.027	0.019	0.160
4-8h awake	3.55	2.08	0.089	-0.008	0.022	0.720
12h+ awake	2.59	1.85	0.161	-0.024	0.019	0.219
Classical $\times$ 4-8h	1.25	2.70	0.643	0.009	0.028	0.741
Hardstyle $\times$ 4-8h	0.73	2.70	0.787	0.055	0.028	0.053
Classical $\times$ 12h+	-0.67	2.47	0.788	0.007	0.026	0.788
Hardstyle $\times$ 12h+	0.53	2.51	0.834	0.020	0.026	0.442
Group Var	575.41	66.74		0.001	0.022	

Note: Coefficients (Coef.), standard errors (SE), and *p*-values are shown for each predictor.

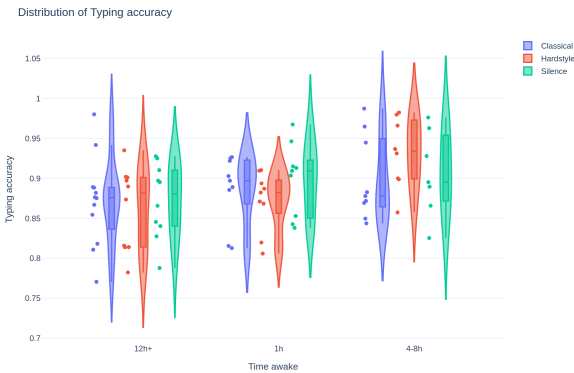


Figure 1: Typing accuracy by time awake (grouped by music type).



Figure 2: Typing accuracy by music type (grouped by time awake).



Figure 3: Typing speed (WPM) by time awake (grouped by music type).



Figure 4: Typing speed (WPM) by music type (grouped by time awake).



Figure 5: Interaction plot for typing speed (WPM) by music type and time awake.

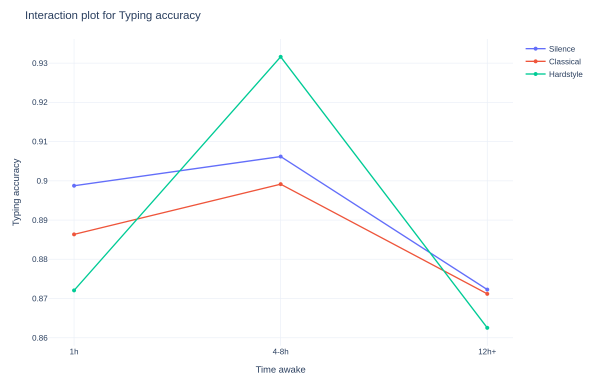


Figure 6: Interaction plot for typing accuracy by music type and time awake.

Table 3: Tukey HSD post hoc comparisons for typing speed (WPM) between all group pairs. No comparisons were statistically significant ($p > 0.05$).

Group 1	Group 2	Mean Diff.	<i>p</i> -adj	Significant
Classical_12h+	Classical_1h	12.78	0.902	False
Classical_12h+	Classical_4-8h	20.89	0.382	False
Classical_12h+	Hardstyle_12h+	2.36	1.000	False
Classical_12h+	Hardstyle_1h	13.69	0.862	False
Classical_12h+	Hardstyle_4-8h	21.53	0.388	False
Classical_12h+	Silence_12h+	-0.74	1.000	False
Classical_12h+	Silence_1h	13.58	0.867	False
Classical_12h+	Silence_4-8h	18.69	0.635	False

Note: Only the most significant findings were included in the table.

Table 4: Tukey HSD post hoc comparisons for typing accuracy between all group pairs. No comparisons were statistically significant ($p > 0.05$).

Group 1	Group 2	Mean Diff.	p -adj	Significant
Classical_12h+	Classical_1h	0.015	0.999	False
Classical_12h+	Classical_4-8h	0.028	0.925	False
Classical_12h+	Hardstyle_12h+	-0.009	1.000	False
Classical_12h+	Hardstyle_1h	0.001	1.000	False
Classical_12h+	Hardstyle_4-8h	0.060	0.153	False
Classical_12h+	Silence_12h+	0.001	1.000	False
Classical_12h+	Silence_1h	0.028	0.930	False
Classical_12h+	Silence_4-8h	0.035	0.842	False

Note: Only the most significant findings were included in the table.

4 Discussion

The present pilot study found *no reliable evidence* that either background-music tempo or time awake modulates short-duration typing performance. The descriptive peak at 4-8h awake aligns with the homeostatic-circadian model, which predicts optimal alertness several hours after sleep onset [Borbély, 1982]. However, the mixed-effects analysis indicates that the observed differences are well within sampling error.

Three methodological factors likely constrained sensitivity:

1. **Limited replication.** Only four participants completed the full nine-session schedule, reducing the balanced information that repeated-measures designs normally provide.
2. **Self-reported wake time.** Misreporting or clock-reading errors could have blurred the time-awake factor.
3. **Learning and fatigue.** The stimulus list was short and repeatedly sampled, so practice and boredom effects may have counteracted the experimental manipulations.

5 Conclusion

This pilot study found no evidence that instrumental background music—whether slow classical or fast hard-style—affects typing speed or accuracy, nor that music moderates the modest circadian pattern observed across the waking day. These results they should be viewed in light of three constraints: a small convenience sample, partial completion of the planned repeated-measures schedule, and reliance on self-reported wake duration.

Future experiments should include a larger sample group, fully compliant participants. Ideally more scientific methods for sleep tracking, and not self-reporting. Until experiments with stricter surveillance, and preferably more resources; you can listen to whatever music you like.

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